

Cambridge (CIE) IGCSE Chemistry
Paper 2: Multiple Choice (Set F)
Extended

Saturday 26 April 2025

Morning (Time: 0 hours 45 minutes)

Total marks

/ 40

Instructions

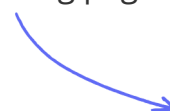
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.

Materials

- A copy of [the Periodic Table of elements](#) is required for this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



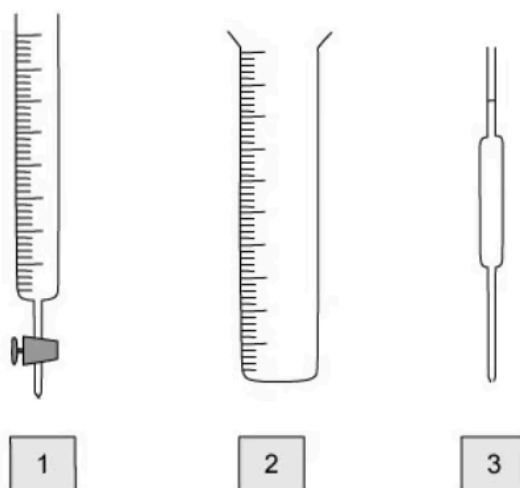
1 Four substances are shown below.

In which of these substances are the particles close together and moving slowly past each other?

- A. Ice
- B. Air
- C. Steam
- D. Water

(1 mark)

2 Measuring volumes of liquids is a fundamental aspect of experimental chemistry. The diagram shows three different pieces of apparatus used for this task.



Which row is correct?

	1	2	3
A	burette	measuring cylinder	pipette
B	burette	pipette	measuring cylinder
C	pipette	measuring cylinder	burette
D	measuring cylinder	burette	pipette

(1 mark)

3 Pure H₂O boils at 100 °C and freezes at 0 °C.

What is the melting point and boiling point of water after the addition of potassium chloride?

	Boiling point / °C	Freezing point / °C
A	103	-3
B	103	3
C	97	-3
D	97	No change

(1 mark)

4 Element X contains 4 electrons in its outer shell and 125 neutrons in its nucleus. What is element X?

- A. Beryllium
- B. Carbon
- C. Tin
- D. Lead

(1 mark)

5 Which statement about graphite and diamond is not correct?

- A. Graphite is a good lubricant due to the weak forces in between the layers.
- B. Diamond is used as a drill tip because it is an extremely hard substance.
- C. Graphite has a low melting point due to the weak forces in between the layers.
- D. Diamond is an allotrope of carbon.

(1 mark)

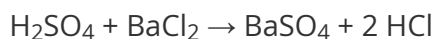
6 CH₄, H₂O and HCl are covalent compounds.

Which atoms in these compounds do not use all of their outer shell electrons in bonding?

- A. C and O
- B. C and H
- C. Cl and O
- D. Cl and C

(1 mark)

7 Dilute sulfuric acid and barium chloride solution react to produce barium sulfate and hydrochloric acid, as shown in the reaction below:

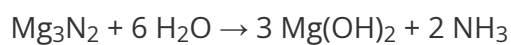


Which row represents the correct state symbols for the reaction?

	H₂SO₄	BaCl₂	BaSO₄	HCl
A	(aq)	(l)	(s)	(aq)
B	(l)	(aq)	(aq)	(l)
C	(aq)	(aq)	(s)	(aq)
D	(l)	(aq)	(l)	(aq)

(1 mark)

- 8 Magnesium nitride and water react to produce a precipitate of magnesium hydroxide and ammonia gas.



How much magnesium nitride is needed to produce 87 g of the precipitate in excess water?

- A.** 24 g
- B.** 50 g
- C.** 30 g
- D.** 12 g

(1 mark)

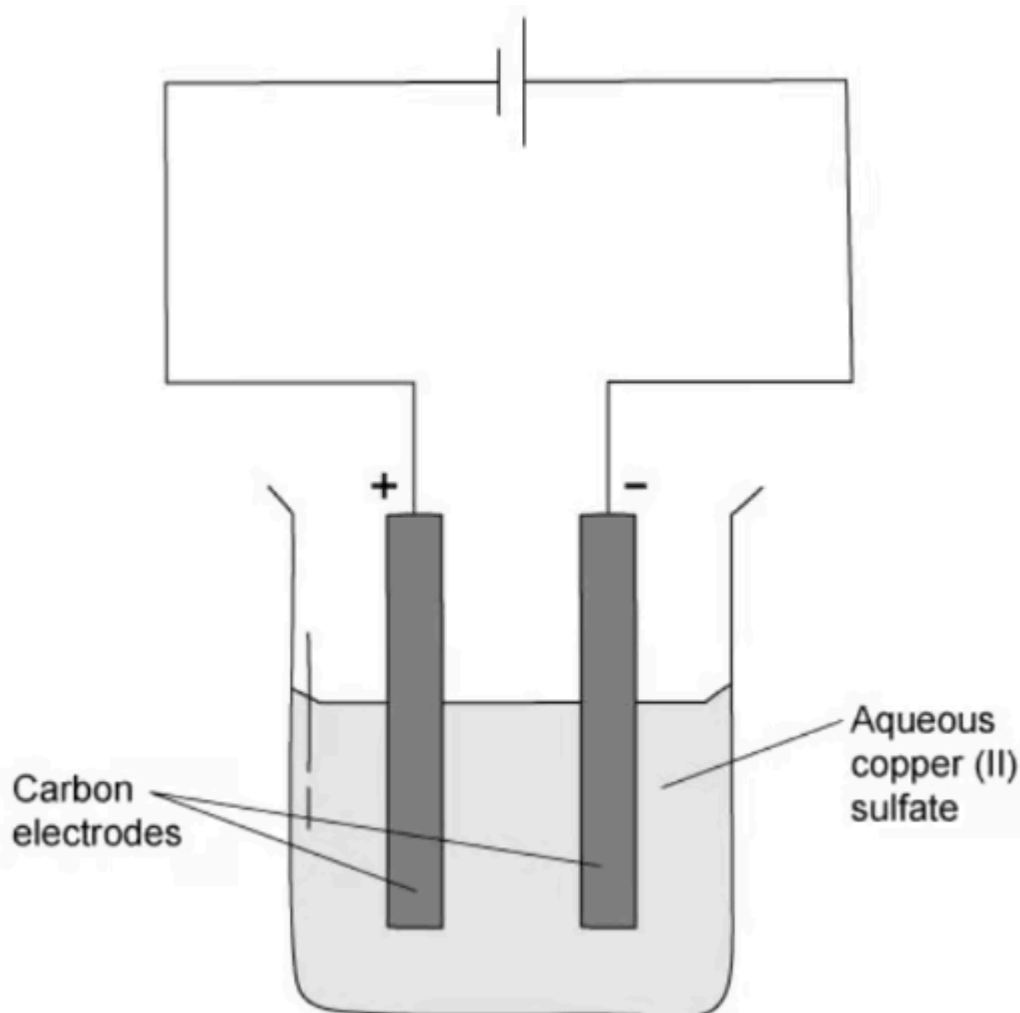
- 9 6 g of hydrogen react with 48 g of oxygen to form water. A chemist performs the same reaction using 2 g of hydrogen.

What mass of oxygen is required to combine with the hydrogen?

- A. 24 g
- B. 12 g
- C. 16 g
- D. 72 g

(1 mark)

- 10 The diagram shows the electrolysis of aqueous copper(II) sulfate.

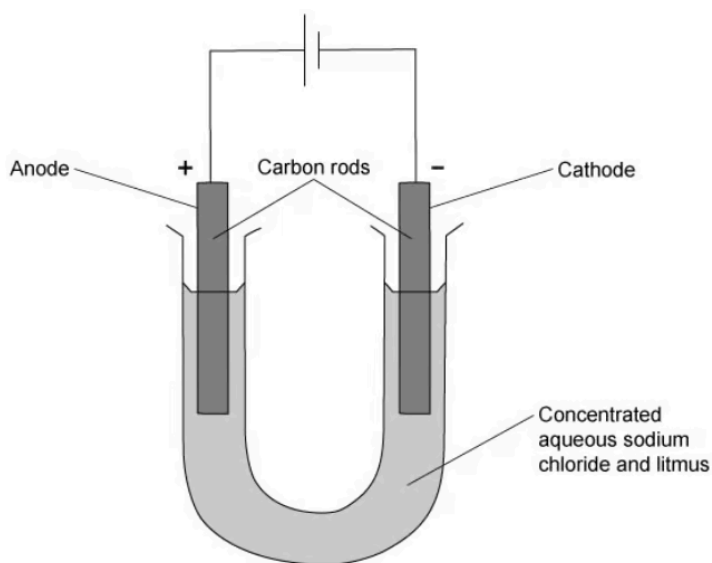


Which statement is correct?

- A.** Copper metal is deposited at the positive electrode.
- B.** In the external circuit the electrons move from negative to positive.
- C.** In the solution the electrons move from positive to negative.
- D.** Oxygen gas is produced at the positive electrode.

(1 mark)

11 The diagram shows the electrolysis of concentrated aqueous sodium chloride.



What is the colour of the litmus at each electrode after five minutes?

	colour at anode	colour at cathode
A	blue	red
B	red	blue
C	red	colourless
D	colourless	blue

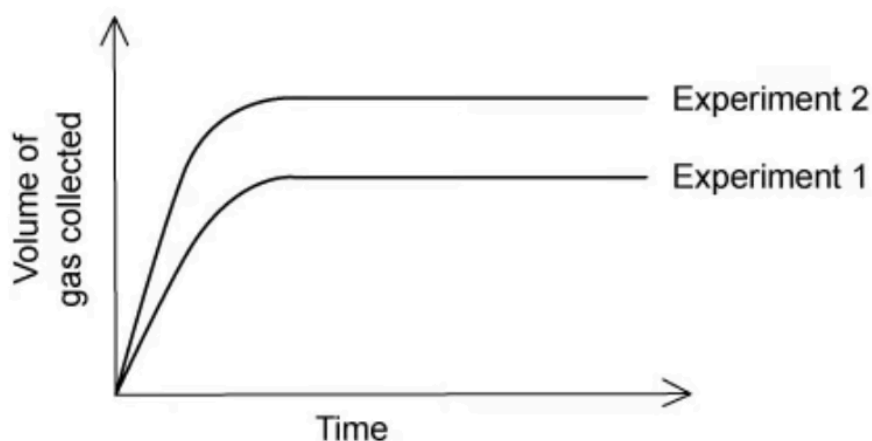
(1 mark)

12 Which of the following processes represents a physical change?

- A. Acid base neutralisation
- B. Metal displacement reactions
- C. Boiling ethanol
- D. Combustion of magnesium

(1 mark)

13 The graph shows the results of two separate reaction rate investigations using excess calcium carbonate and sulfuric acid.



Which statement explains the difference between the results?

- A. The temperature was higher in experiment 2.
- B. Experiment 2 was allowed more time to react.
- C. Same volume of a more concentrated acid was used in experiment 2.
- D. More calcium carbonate was used in experiment 2.

(1 mark)

14 Which indicator displays the correct colour in each solution?

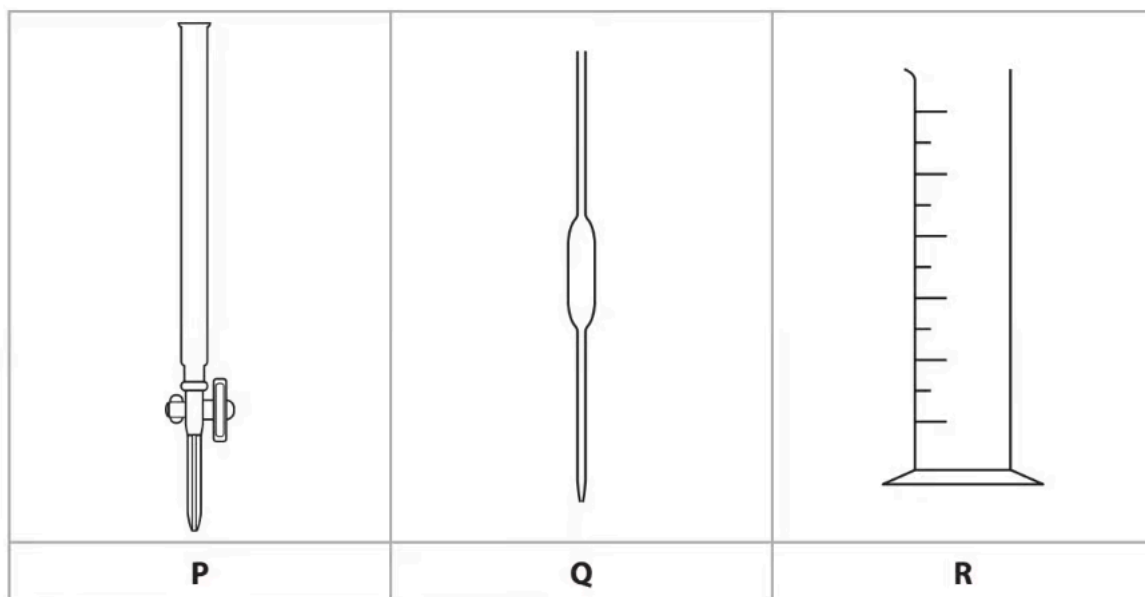
Indicator	Colour in dilute sulfuric acid	Colour in potassium hydroxide solution
methyl orange	red	yellow
universal indicator	red	orange
litmus	red	purple
thymolphthalein	blue	colourless

- A.** Methyl orange
- B.** Universal indicator
- C.** Litmus
- D.** Thymolphthalein

(1 mark)

15 Different salts can be prepared by titrating an alkali with an aqueous acid.

Three pieces of apparatus, **P**, **Q** and **R**, used to measure volumes of liquid are shown below.



What are the names of the three pieces of apparatus, **P**, **Q** and **R**?

	P	Q	R
A	pipette	burette	measuring cylinder
B	burette	pipette	conical flask
C	pipette	volumetric flask	burette
D	burette	pipette	measuring cylinder

(1 mark)

16 Endothermic reactions can be used in emergency sports injury treatment packs.

What is true about these packs?

- A.** Absorb energy from the surroundings and transfer it to the injury
- B.** Can be re-used
- C.** Usually contain ammonium nitrate and water in separate compartments
- D.** Cause the temperature of the surroundings to increase

(1 mark)

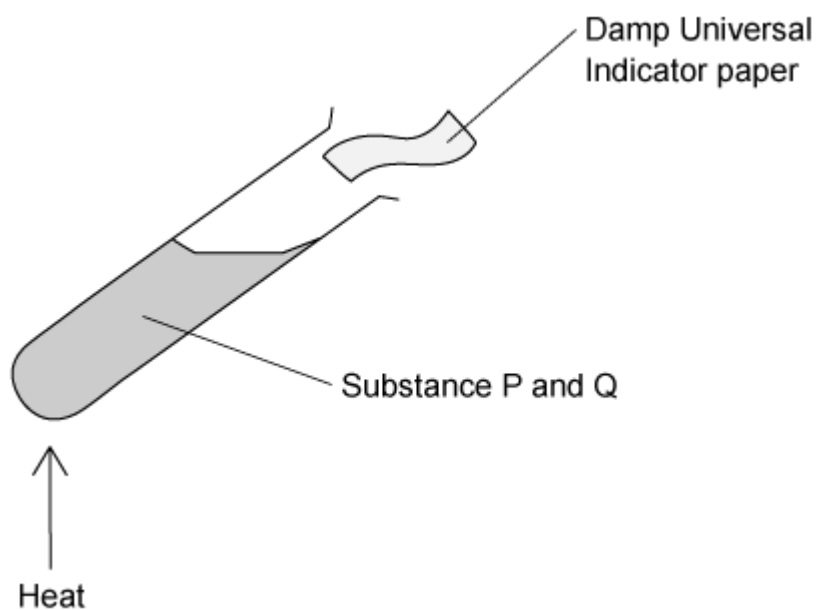
17 A student wishes to electroplate an object with copper.

Which row is correct?

	object is made at the	a suitable electrolyte is
A	anode	CuO(s)
B	anode	CuSO ₄ (aq)
C	cathode	CuO(s)
D	cathode	CuSO ₄ (aq)

(1 mark)

18 A student is testing two substances, **P** and **Q**, as shown.



The results of the test are that the Universal indicator paper turns blue. What are substances **P** and **Q**?

- A. HCl and NH_4NO_3
- B. NaOH and NH_4NO_3
- C. NaCO_3 and HCl
- D. NaCO_3 and NaOH

(1 mark)

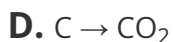
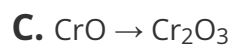
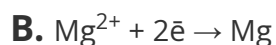
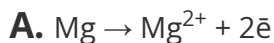
19 A substance reacts with dilute H_2SO_4 to form a salt which is separated by filtration.

What is the substance?

- A. Aqueous sodium hydroxide
- B. Aqueous barium chloride
- C. Copper
- D. Copper(II) carbonate

(1 mark)

20 In which reaction is the reactant being reduced?



(1 mark)

21 Which statement about the Alkali metals is correct?

A. Potassium is more dense than lithium.

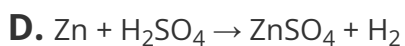
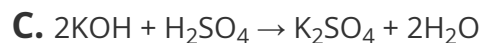
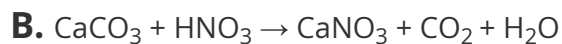
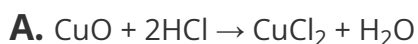
B. Lithium has a higher density than sodium.

C. Potassium is less reactive than sodium.

D. The melting point of sodium is higher than lithium.

(1 mark)

22 Which equation does **not** show the correct reaction of an acid?



(1 mark)

23 Element X is an unreactive non-metal element. Where could this element be found on the Periodic table?

A. At the far left hand side

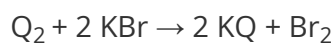
B. At the far right hand side

C. At the top of Group VI

D. At the bottom of Group VII

(1 mark)

- 24** Element Q is a halogen which reacts with aqueous potassium bromide, as shown in the reaction below:



Which row describes element Q?

	More reactive than bromine	Greater A_r than bromine
A	✓	✓
B	✓	×
C	×	✓
D	×	×

(1 mark)

- 25** The following section of the reactivity series shows a newly discovered element, Y. The only oxide of Y has the formula YO.

Ca

Mg

Fe

Y

H

Cu

Which of the equations shows a reaction which would take place?

- A.** $2\text{Y(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow 2\text{Y}^+(\text{aq}) + \text{Cu(s)}$
- B.** $\text{Y(s)} + \text{Fe}_2\text{O}_3(\text{s}) \rightarrow 2\text{Fe(s)} + 3\text{YO(s)}$
- C.** $\text{Y(s)} + 2\text{HCl(aq)} \rightarrow \text{YCl}_2(\text{aq}) + \text{H}_2(\text{g})$
- D.** $\text{Cu(s)} + \text{Y}^{2+}(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Y(s)}$

(1 mark)

- 26** The elements of the Periodic table can be broadly classified into two main characters which are metals and nonmetals. Which statements are correct?

1	Magnesium is more metallic than silicon.
2	Boron is more metallic than lithium.
3	Carbon is less metallic than beryllium.
4	Sodium is less metallic than aluminium.

- A.** 1 and 3

- B.** 1 and 2
- C.** 2, 3 and 4
- D.** 1, 2 and 4

(1 mark)

27 Why are alloys harder than the pure metals from which they are made?

- A.** There are stronger bonds between the atoms
- B.** The atoms are all the same size
- C.** The layers of atoms are distorted
- D.** The properties of each element have combined

(1 mark)

28 A student analysed a solution and found it to contain ions of silver and barium. What could the identity of the anion be?

- A.** Nitrate
- B.** Sulfate
- C.** Chloride
- D.** Carbonate

(1 mark)

29 Aluminium is extracted from its oxide by electrolysis.

The oxide is dissolved in**1**..... cryolite and aluminium is deposited at the**2**.....

Which words correctly complete gaps 1 and 2?

	1	2
A	aqueous	cathode
B	aqueous	anode
C	molten	cathode
D	molten	anode

- A.** Option A
- B.** Option B
- C.** Option C
- D.** Option D

(1 mark)

- 30** A student prepared four iron nails as explained below in an experiment to investigate rusting. She allowed the nails to rest in individual test tubes in a laboratory for one week.

Which nail will show the greatest amount of rust?

- A.** Dipped in water and allowed to dry.
- B.** Dipped in paint and allowed to dry.
- C.** Coated in grease.
- D.** Galvanised with zinc.

(1 mark)

31 Mineral water contains dissolved salts such as magnesium chloride. Which one of the following statements about mineral water is correct?

- A.** Mineral water boils at slightly above 100 °C.
- B.** Mineral water is pure water.
- C.** Mineral water boils at exactly 100 °C.
- D.** Another name for mineral water is fizzy water.

(1 mark)

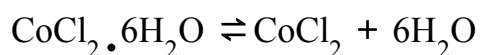
32 Petrol engines produce pollutants as a by-product of internal combustion. Catalytic converters are fitted to cars to reduce the level of pollution they produce in their exhaust gases.

Which statement about this process is **not** correct?

- A.** Nitrogen and oxygen react to produce nitrogen dioxide.
- B.** Carbon monoxide undergoes oxidation in the catalytic converter.
- C.** Carbon monoxide is formed when petrol undergoes complete combustion.
- D.** Nitrogen monoxide undergoes reduction in the catalytic converter.

(1 mark)

33 Hydrated cobalt(II) chloride undergoes thermal decomposition when heated to produce anhydrous cobalt(II) chloride and water.



The addition of water to CoCl_2 rehydrates the compound. Which row correctly describes the reverse reaction?

	reaction type	colour change
A	exothermic	blue to pink
B	endothermic	blue to pink
C	exothermic	pink to blue
D	endothermic	pink to blue

(1 mark)

34 What are the products at each electrode in the extraction of aluminium?

	cathode	anode
A	oxygen	aluminium
B	chlorine	aluminium
C	aluminium	chlorine
D	aluminium	oxygen

A. Option A

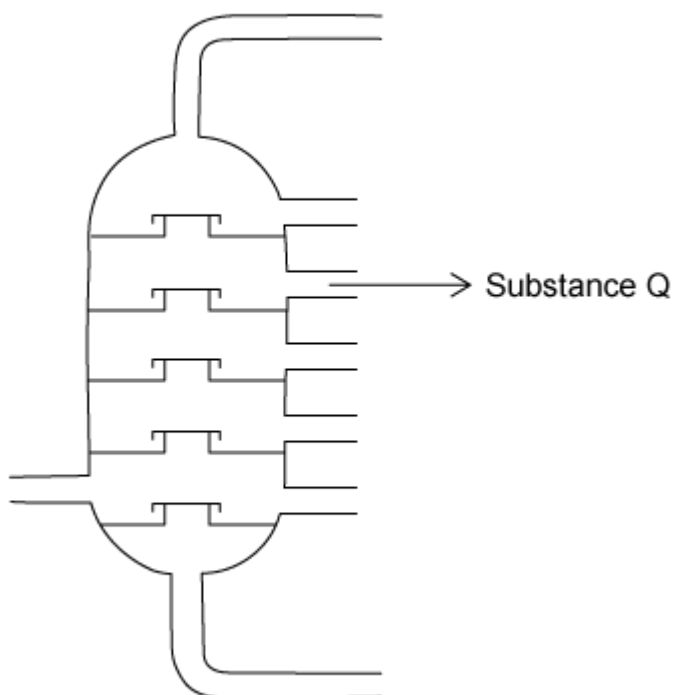
B. Option B

C. Option C

D. Option D

(1 mark)

- 35** The diagram shows an industrial process involved in the production of a substance that is used as an aircraft fuel.



	process	substance Q
A	fractional distillation	petrol
B	thermal decomposition	paraffin
C	fractional distillation	paraffin
D	thermal decomposition	petrol

(1 mark)

- 36** Ethanol is used as a solvent and fuel, it is produced by the1..... of2..... .

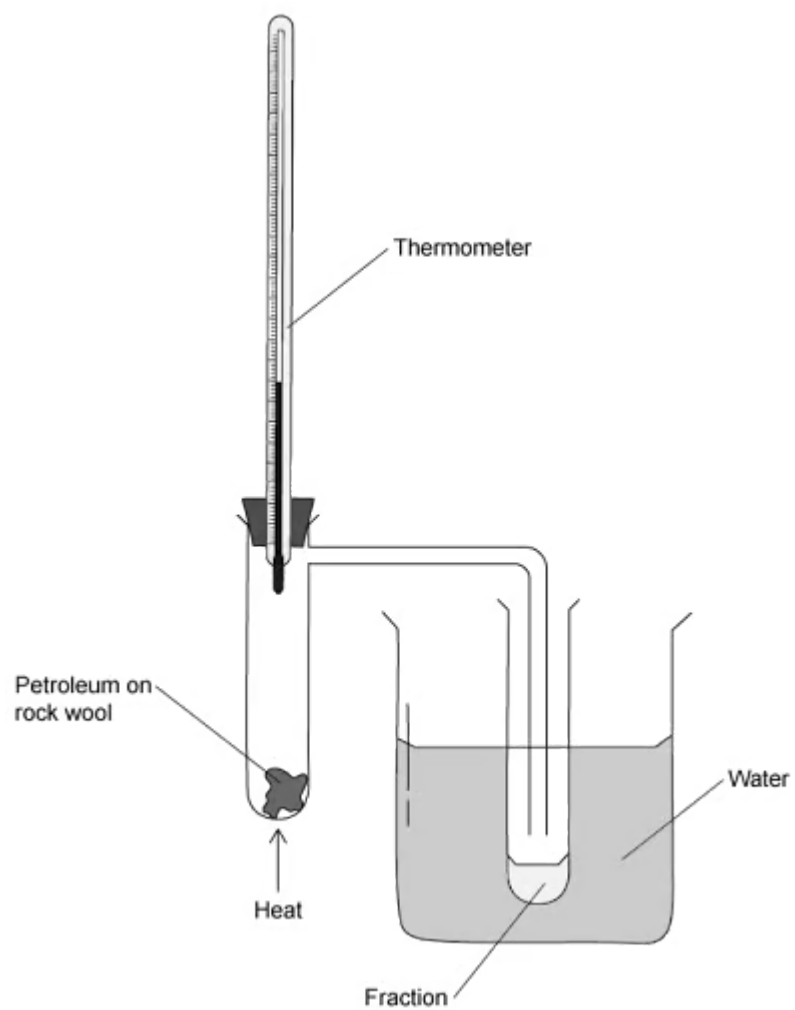
Which words would correctly complete the gaps 1 and 2?

	1	2
A	combustion	ethane
B	combustion	glucose
C	fermentation	glucose
D	fermentation	ethane

(1 mark)

37 The diagram below shows the apparatus used to separate petroleum (crude oil) into four

fractions.



Which fraction contains the smallest hydrocarbon molecules?

fraction	boiling point range / °C
A	up to 70
B	71 to 120
C	121 to 170
D	over 171

(1 mark)

- 38** When the number of atoms in a molecule of a hydrocarbon increases, the amount of energy released when it burns increases.

What is the correct order?

	less energy	→	more energy released
A	methane	ethene	ethane
B	ethene	ethane	methane
C	ethene	methane	ethane
D	methane	ethane	ethene

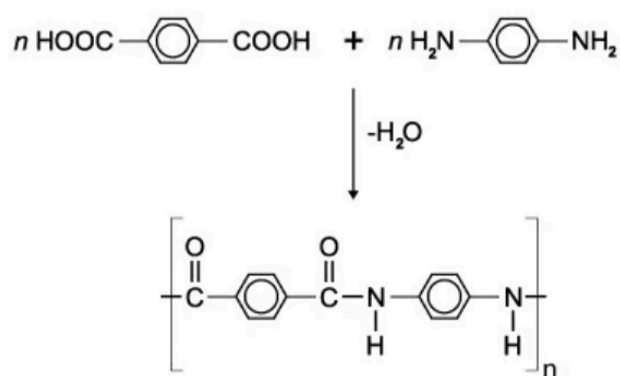
(1 mark)

39 Which of these statements about carboxylic acids is **not** correct?

- A. Aqueous ethanoic acid has a pH less than 7.
- B. Methyl orange turns yellow in aqueous ethanoic acid.
- C. They produce carbon dioxide when reacted with a metal carbonate.
- D. They contain the functional group -COOH.

(1 mark)

40 The equation shows the formation of a polymer called Kevlar.



Which row describes Kevlar?

	how the monomer is formed	type of polymer
A	addition polymerisation	polyamide
B	addition polymerisation	polyester
C	condensation polymerisation	polyamide
D	condensation polymerisation	polyester

(1 mark)